

Arsitha Sathu

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SUMMARY

Software Engineer with hands-on experience building production-grade systems using Java, Spring Boot, React, Node.js, and cloud services on Azure and AWS. I've developed scalable APIs, real-time voice-bot workflows, and microservices that improve system reliability and performance. Known for clean code, strong problem-solving, and owning features end-to-end while delivering practical engineering results.

EDUCATION

- **George Mason University** Fairfax, VA, USA
Masters in Computer Science; GPA: 3.83 *Aug 2023 – May 2025*
- **Jawaharlal Technological University** Telangana, India
Bachelors in Computer Science; CGPA: 9.21 *Aug 2017 – June 2021*

EXPERIENCE

- **Blue Horse Digital** Richmond, VA
Software Engineer *July 2025 - Present*
 - Built the core speech-to-speech Voicebot workflow using Azure Container Apps, Azure AI Foundry (GPT-4o Realtime), and Azure Key Vault.
 - Developed backend services in FastAPI and integrated Azure Speech SDK for smooth STT/TTS interactions.
 - Implemented WebSocket communication to support low-latency, real-time audio exchange with the GPT-4o model.
 - Automated CI/CD pipelines to build and deploy containerized services through Azure Container Registry and Azure Container Apps.
 - Centralized secret management with Azure Key Vault, strengthening security and ensuring consistent, safe deployments.
- **Infosys Limited** Hyderabad, Telangana, India
Fullstack Developer *Oct 2021 - July 2023*
 - Built fullstack features using Java, Spring Boot, Angular, and TypeScript, developing REST APIs, UI components, and database logic for production systems.
 - Designed and optimized microservices integrated with PostgreSQL, improving API performance by 20–30% through query tuning and refactoring slow endpoints.
 - Created reusable frontend components (dynamic tables, filters, role-based views) that reduced duplicated code and standardized dashboards across teams.
 - Debugged and resolved production issues by analyzing logs, reproducing failures, and implementing fixes across backend and frontend layers.
 - Ensured maintainable, high-quality code by applying SonarQube standards and writing JUnit tests, consistently achieving more than 95% coverage.

PROJECTS

- **AI Log Analysis:**
GitHub: <https://github.com/arsitha20/AI-Log-Analysis-Dashboard>
 - Built a full-stack AI log analysis system where Spring Boot ingests, parses, and stores raw logs in PostgreSQL, then calls a FastAPI microservice that uses GPT-4o-mini to cluster errors, identify root causes, and generate debugging insights.
 - Developed an interactive React dashboard that supports one-click log analysis, visualizes error clusters, and displays structured explanations and recommended fixes, making it easier to spot recurring failures across distributed services.
- **Abstractive text summarization:**
GitHub: <https://github.com/arsitha20/CS678-Abstractive-Text-Summarization-Telugu>

- Fine-tuned an mT5 transformer on a custom-curated Telugu narrative dataset to generate coherent abstractive summaries, addressing linguistic challenges such as free-word order and multi-sentence compression.
- Evaluated model quality using ROUGE-1/2/L metrics and implemented preprocessing pipelines for tokenization, text normalization, and dataset balancing to improve summary fluency and content retention.
- **Departmental Connect:**
GitHub: <https://github.com/arsitha28/CSE-Department-Website>
 - Redesigned a static departmental website into a dynamic React and Node.js platform with searchable course content, faculty pages, announcements, and admin-friendly content updates backed by SQL.
 - Improved site usability with responsive UI components, optimized data queries, and modularized backend endpoints, enabling faster navigation and easier content management for students and faculty.

SKILLS

- **Languages:** Java, Python, SQL, C
- **Backend:** Spring Boot, Node.js, REST APIs, GraphQL, Microservices, Maven
- **Frontend:** React.js, Angular, HTML, CSS, TypeScript, JavaScript, JSON
- **Database:** PostgreSQL, MySQL, MongoDB, SQL
- **Devops & Cloud:** CI/CD Pipelines, Docker, Jenkins, Git, Bitbucket, AWS (EC2, S3), Azure (AI Services, Container Apps, Key Vault)
- **Testing:** JUnit, Jest, Postman, Swagger, SonarQube
- **Tools & Scripting:** Bash, Shell, Linux, VS Code, IntelliJ, Postman
- **Coursework:** Data Structures & Algorithms, Operating Systems, OOPS Concepts, DBMS, Web Development, Software Engineering, Data Mining, NLP, LLMs

ACHIEVEMENTS AND RESPONSIBILITIES

- **1st position:** In Webathon, CSE Department Technical Competition. Webathon was a department-wide web design contest where teams built a complete website within a limited timeframe. Our team secured 1st place and was recognized as the Best Web Designer.
- Received Outstanding Academic Achievement Award for academic excellence during MS in Computer Science at George Mason University.
- Department Magazine Coordinator and Class Representative for 4 consecutive years during my undergrad.